



Fig. 2: Typical pacemaker (Courtesy: www.bostonscientific.com)



Fig. 3: MPX2300DT1 pressure sensor from Freescale Semiconductor

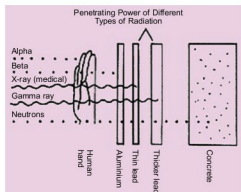


Fig. 4: Penetration power of different types of radiation (Courtesy: www.ratical.org)

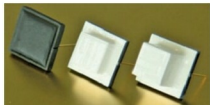


Fig. 5: X100-7 gamma radiation detector (Courtesy: www.first-sensor.com)

electrodes of pacemakers detect the heart's electrical activity and send data through wires to the computer. So, these are life-saving and safety-critical medical applications of sensors.

Device malfunctions exist and it's often difficult to pinpoint the exact

cause. For example, if a pacemaker fails, the reason could be equipment failure or changes in the patient's native heart rhythm. Sometimes inaccurate data due to incorrect placement of the device on the body could seriously affect the results of diagnosis. Errors can be significantly reduced with the right approach and good-quality product design. Sensors for medical

devices must comply with safety, accuracy and reliability standards laid down by the regulatory bodies.

For example, IEC 60601-1 medical device safety standard establishes general requirements for safety and performance and is the global benchmark for medical electrical equipment. Apart from IEC 60601, statutory specifications for medical devices include other enforcement and standards for quality management, risk management, usability and functional safety to ensure that a given apparatus functions correctly in response to inputs.

Good-quality sensors complying with safety standards provide greater accuracy and reliability to patient care. For instance, high-volume pressure sensors from Freescale Semiconductors and airflow sensors from Honeywell are intended for use in medical devices.

Freescale's MPX2300DT1 high-volume pressure sensor with integrated thin-film temperature compensation and calibration is based on piezoresistive technology. It is intended for use in medical diagnostics, infusion pumps, blood pressure monitors, pressure catheter applications and patient monitoring. It can be interfaced with an embedded system.

Honeywell's Zephyr digital airflow sensors featuring precise measurement, fast response time, high accuracy, high stability and high sensitivity are suitable for critical medical applications like anesthesia delivery machines, laparoscopy and heart pumps. These support ASIC-based I2C digital output, which allows easy integration to microprocessors or microcontrollers.

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